

Fall 2023 Math 8600 Mathematical Foundations of Deep Learning

Instructor

Dr. Xiaojing Ye (E-mail: xye@gsu.edu, Web: <https://math.gsu.edu/xye>).

Office Hours and Location

TR 8:00–9:15 AM or by appointment. 25 Park Place Room 1436.

Class Time and Location

TR 9:30–10:45 AM, Langdale Hall 525. Lectures are also live-streamed and recorded in WebEx.

Lecture Notes and Textbook

The course materials are covered by the instructor's lecture notes. There is no textbook for this course. Additional references (tutorials and related research papers) will be provided.

Prerequisites

Knowledge in Optimization (MATH 4211/6211) and experience in Python coding are helpful but not required. If you need an approval to override prerequisite, please fill in the form at https://math.gsu.edu/xye/course/math-dl_handout/prereq_override_form.docx and send the form to me (xye@gsu.edu), Dr. Jason Li (zli@gsu.edu) and Ms. Katina Akins (kakins@gsu.edu).

Course Description

Basic concepts in deep learning theory; Basic structure of standard and convolutional deep neural networks (DNNs); DNN training; Case studies of modern DNN architectures and applications. This course will also introduce concepts on convex and nonconvex analysis, several first-order deterministic and stochastic optimization algorithms, and basics of optimal control and its connection to network design and training.

Course Schedule

1. Basic Concepts of Deep Learning
 - (a) Briefs of deep learning: What is deep learning, approaches, and applications of deep learning.
 - (b) Basic machine learning theory: assessment of accuracy, generalization, under-fitting and overfitting, bias and variance decomposition of prediction error, sample, and computational complexity.
 - (c) Regression and classification in supervised learning: Maximum likelihood estimator, Bayes rule and maximum a priori estimation, linear regression and logistic regression models and computations.
2. Brief Introduction to Deep Neural Networks (DNNs)
 - (a) Artificial neural network (ANN): Components and architecture of ANNs; Universality theory;
 - (b) Convolutional neuron network (CNN): Components and architecture of CNNs, convolutional operations, dropout, batch normalization, basic architectures of CNNs;
 - (c) DNN training: Backward propagation, minimization of loss function, gradient decent algorithm for nonconvex optimization, stochastic gradient decent algorithm, and adaptive stochastic gradient decent algorithms;
 - (d) Case study and applications.
3. Basic Concepts on Convex/Nonconvex Analysis and Optimization Algorithm Inspired DNNs:
 - (a) Basic concept on convex/nonconvex and smooth/nonsmooth optimizations: Definition and basic properties of convex functions, L-smooth function, convex conjugate function; Optimality conditions and global/local solutions.
 - (b) Sub-gradient and sub-differential: Definition, properties, and calculation;
 - (c) Optimization algorithm inspired CNNs: Advantage of algorithm unrolling (or unfolded) DNNs, learnable variational models and algorithms, proximal gradient algorithm and ISTA-net, primal dual algorithm and primal-dual-net, gradient decent algorithm and Variational-net, alternating direction method of multipliers (ADMM) and ADMM-net.
 - (d) Bi-level optimization in meta-learning: Goal and advantage of meta learning, general framework of bi-level optimization for meta-learning, methods for solving bi-level optimization problems
4. Deep learning from optimal control perspective:

- (a) Brief introduction to optimal control: Basic optimal control problem in discrete time and continuous time, Hamiltonian and Pontryagin Maximum Principle (PMP) in discrete and continuous time;
- (b) DNN from dynamical systems perspective: Neural ordinary differential equations for supervised learning, Case study: Neural Ordinary Differential Equations;
- (c) Deep learning from optimal control aspect: First order optimality condition from PMP, numerical methods, Relationships between back-propagation and Hamiltonian maximization and method of Lagrange multipliers for network training, applications in image analysis;

Attendance and Administrative Drop Policy

Requirements for class attendance and make-up exams, assignments, and other work in this course are consistent with university policies that can be found at <https://codeofconduct.gsu.edu>.

Course Requirement 1: Final Project

There is final project (50 points) during the semester. The project consists of a deep learning problem of adequate complexity. Successful completion of a project requires theoretical derivations, algorithm development, and numerical implementations and experiments. A completed project must be presented as typeset report in PDF format and a zip file of computer code to reproduce the results presented in the report. Topic and requirements of the project will be given during the semester.

- Project reports and codes are to be submitted electronically on iCollege at <https://icollege.gsu.edu>.
- Late project receives penalty of 10 points every 24 hours after the due time.

Course Requirement 2: Presentation

- There is a final presentation (35 minutes) of each student to be scheduled in the last few weeks of the class. A student should present the theory and numerical results of a research paper. The student can select the paper of his/her own interests with sufficient intellectual depth approved by the instructor or be assigned with a paper by the instructor. Additional instructions will be given in advance.
- All non-presenting students are required to attend and encouraged to ask questions during each presentation.
- The presentation is counted for 50 points of the final grade. It will be evaluated by the instructor based on clarity and intellectual depth of the presentation.

Grading Policy

Projects (50 points) + Final presentation (50 points) = Total (100 pts).

Score	0–59	60–69	70–76	77–79	80–82	83–86	87–89	90–92	93–96	97–100
Grade	F	D	C	C+	B–	B	B+	A–	A	A+

Accommodation

Students who wish to request an accommodation may do so by registering with the Access and Accommodation Center (AACE). Students may only be accommodated upon issuance by the Access and Accommodation Center of a signed Accommodation Plan and are responsible for providing a copy of that plan to instructors of all classes in which accommodations are sought.

Academic Honesty

All materials submitted for grading must be completed independently without help from any other persons or resources (including but not limited to solution manual and online solution websites.) *Cheating and plagiarism are strictly forbidden.* A first occurrence will result in a grade of zero on the assignment for all concerned parties as well as an Academic Dishonesty form being filed with the Dean of Students. A second occurrence will result in a grade of F for the course for the concerned parties and a second Academic Dishonesty form being filed. This course is conducted in a manner consistent with the university policy on academic honesty at <http://codeofconduct.gsu.edu/>.

Electronic Technology Policies

You should not use your cell phone or computer or another device for tasks that are not class related, i.e., you should not text, talk, or do anything that may be considered by your instructor to be disruptive. If you have an emergency, for which using your cell phone is absolutely necessary, quietly step out of the class to handle the emergency.

Alternative Lecture Arrangement Policy

If the University is closed due to inclement weather or health crisis, the lecture and exam will be arranged in an alternative format and announced through iCollege.

Sharing of Instructor-generated Materials

The selling, sharing, publishing, presenting, or distributing of instructor-prepared course lecture notes, videos, audio recordings, or any other instructor-produced materials from any course for any commercial purpose is strictly prohibited unless explicit written permission is granted in advance by the course instructor. This includes posting any materials on websites such as Chegg, Course Hero, OneClass, Stuvia, StuDocu and other similar sites. Unauthorized sale or commercial distribution of such material is a violation of the instructor's intellectual property and the privacy rights of students attending the class, and is prohibited.

COVID-19 Supplements**Supplemental Information for Attendance Policies**

Students who want to do well in this course will attend class following the class attendance policy. You will need an excused absence due to illness. GSU has a new process for students seeking excused absences through the Dean of Students Office. Please submit documentation to

<https://deanofstudents.gsu.edu/student-assistance/professor-absence-notification/>

I will then be notified by the Dean of Students of any excused absences.

Should a student test COVID positive, any accommodations to the class attendance policy will be informed by evolving guidance from the CDC on quarantine. In most cases there will be no major change to mode of course delivery, so students will be responsible for collecting notes for missed classes and making up any work they miss during quarantine. Anyone who has a positive COVID test is encouraged to alert the university so that appropriate contact tracing can be conducted.

Wearing Masks in Class

You are encouraged to wear a face covering in all class meetings. I know that face masks may make some aspects of class more difficult. It will be harder for us all to project our voices and read each other's facial expressions. However, I am willing to sacrifice these elements since wearing a mask is one thing I can control to support the health and safety of our community. Be aware that wearing face mask is not required by GSU, so there is no penalty if you choose to not wear a mask. Our university community has a strong tradition of upholding the value of mutual respect, we therefore ask students to not engage in behavior that would be disruptive if your fellow students make a different choice about wearing masks. If you have concerns, please discuss them with me and I will work to the best of my ability to provide a comfortable environment conducive to student learning.